et al. [Yamamori 26] instead shaped the physical actuator to match the simulation. A commercial valve and pump cannot be redesigned in that way, so this thesis learns the hydraulic realization while PhysX retains the rigid-body mechanics (Figure 2.3).

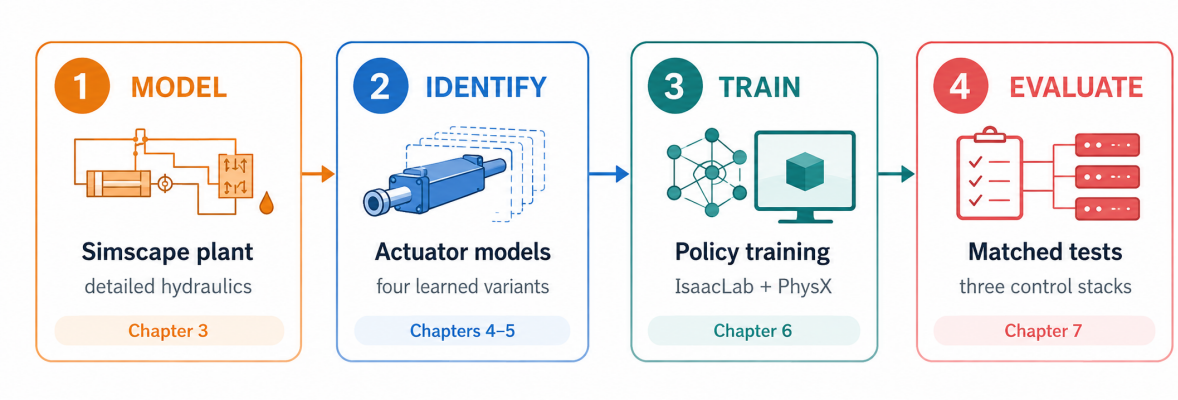


Figure 2.3: Workflow from the hydraulic plant model to matched evaluation. The separation between rigid-body physics and learned actuation follows the actuator-network approach in [Hwangbo 19].

2.6 Domain Randomization and Latent Adaptation

2.6.1 Robustness Through Randomized Training

A model trained on one simulated plant inherits its parameters as assumptions. Domain randomization instead trains across a family, first for visual appearance [Tobin 17] and then for dynamics such as mass, friction, gain and latency [Peng 18]. The hydraulic family in Chapter 4 varies bulk modulus, valve gain, leakage, friction and payload, producing robustness to physically plausible plant variation.

1

2.6.2 From Robustness to Adaptation

Figure 2.4 contrasts a policy trained for one robust response across the plant family with an adaptive stack that conditions its response on the current plant.